

COMP90077

Lecture 3

Submodular Maximization

Caution

- Slides are designed as presentation aid to go along with lecture, not as study material
- Slides not designed to make sense on their own

Today

- Greedy Approximation Algorithms
- Submodular Maximization
- Assignment 1
- Tips

Problems and Instances

- Need to more precisely define problems and instances
- Def. A *combinatorial optimization problem* Π is a collection of instances I . An *instance* consists of:
 - Finite set of elements E
 - **(Objective function)** g that maps subsets of E to real numbers
 - **(Feasible sets/solutions)** Collection $\mathcal{F}$ of subsets of E
 - **Goal:** Find a feasible subset S that maximizes/minimizes g
 - Use $OPT(I)$ to denote the maximum/minimum objective value
- Def. An efficient exact algorithm for Π is one that on every instance I , - outputs a feasible solution S with $g(S) = OPT(I)$ in time polynomial in representation of I

Reductions

Def. Problem Π reduces to problem Π' if for every instance I of Π ,

- can efficiently compute an instance $\phi(I)$ such that
- For every optimal solution S' of $\phi(I)$, can efficiently compute an optimal solution S of I

Approximation Algorithms

- Def. An efficient exact algorithm for Π is one that on every instance I , - outputs a feasible solution S with $g(S) = OPT(I)$ in time polynomial in representation of I
- Many problems are NP-hard
- Widely believed not to have efficient exact algorithms
- Def. A *c-approximation algorithm* for Π is one that on every instance I :
 - runs in polynomial time
 - (maximization) outputs a feasible solution S with $g(S) \leq c \cdot OPT(I)$
 - (minimization) outputs a feasible solution S with $g(S) \geq c \cdot OPT(I)$

Set Systems and Coverage

Def. A *set system* consists of

- A *universe* U of n elements
- A collection of m sets $S_1, \dots, S_m$

Def. Given $X \subseteq [m] = \{1, \dots, m\}$, the *coverage* of X is the number of elements covered by X

$$\text{cov}(X) = \left| \bigcup_{j \in X} S_j \right|$$

Coverage Maximization Problem

Def. An instance of the Coverage Maximization Problem consists of:

- A set system with n elements and m sets $S_1, \dots, S_m$.
- A bound k .
- Goal: find k sets with maximum coverage
- Equivalently, find $I \subseteq [m]$ with $|I| \leq k$ that maximizes $cov(I)$

Coverage Maximization Applications

- Communications tower placement
- Guard placement
- Influence maximization

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- Greedy algorithm?

Greedy Coverage Maximization

- Initialize $I \leftarrow \emptyset$
- While $|I| < k$:
 - Find j^* that maximizes increase in coverage: $\text{cov}(I \cup \{j^*\}) - \text{cov}(I)$
 - Add j^* to I
- Return I

Greedy Analysis

Coverage Maximization Summary

- Greedy achieves a $(1 - 1/e)$ -approximation
- For every constant $\epsilon > 0$, NP-hard to get $(1 - 1/e + \epsilon)$ -approximation.
Implies NP-hard to get $(2/3)$ -approximation
- See why next lecture
- Next: Can we generalize?

Submodular Functions

A function $g: 2^E \rightarrow R$ is *submodular* if

(Diminishing Marginals) For every $S \subseteq T \subseteq E$ and $e \notin T$,

$$g(S \cup \{e\}) - g(S) \geq g(T \cup \{e\}) - g(T)$$

(Definition 2) For every $S, T \subseteq E$,

$$g(S) + g(T) \geq g(S \cup T) + g(S \cap T)$$

Submodular Functions are Everywhere

- Coverage function
- Matroid rank function
- Purchasing cost with buy-at-bulk discounts
- Utility from a set of substitute goods

Cardinality-Constrained Submod. Max.

Def. An instance consists of:

- A non-negative monotone submodular function $g: 2^E \rightarrow R$
- A bound k .
- Goal: find $I \subseteq E$ with $|I| \leq k$ that maximizes $g(I)$
- Greedy algorithm?

Greedy Submodular Maximization

- Initialize $I \leftarrow \emptyset$
- While $|I| < k$:
 - Find j^* that maximizes increase in objective: $g(I \cup \{j^*\}) - g(I)$
 - Add j^* to I
- Return I

Greedy Analysis